Run report: an invariant guess on the second link at $\ell = 8$

- **Study:** `research/dreg_second_link_guess_20261105/`, pre-registered before any L3 run. No amendments.
- **Run:** 2026-11-05 16:42 UTC to 22:33 UTC.
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.
- **Standing:** unverified assertion until a pcc verdict exists.

What ran

- **Arm L3:** `x15 := x7 + c, x44 := h + Σ_{t∈T20} x_{16+t}` (coordinate 20 of `e1 + x3`), `x24 = g`, with 8 branches, on the 12 committed unsatisfiable (22, 8) draws.
  - It runs a new `dreg_linear_guess` build with multi-source `--sum` and `--link2`, at `--to 4 --max-cols 140000 --check`.
  - Early stop at the first branch that saturates degree 4 without refuting.
- **Comparator S3:** the committed records of the two earlier studies.
- **Identity control N:** S3 on draw 5 branch 0, re-run with the new build.
- **Limits:** `prlimit --as=15e9`, `timeout 7200`, one process at a time, on the 4-core, 16 GB container, not isolated.

Completion

- 74 processes started; 73 exited 0.
- One run, L3 on draw 11 branch 7, was killed by a container restart with empty output. That attempt is kept in `runs/attempts/`, and the retry completed.
- Every L3 system has 43 occurring variables. The degree-4 scan took 147–351 s per run.

Measured values

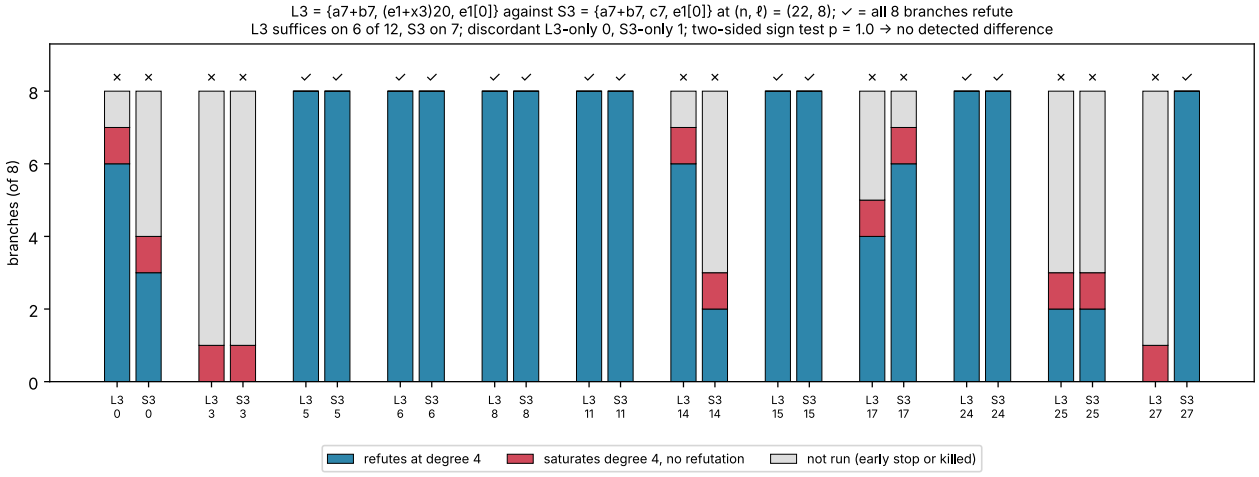


Figure 1: Branches per arm and draw: refuting at degree 4, saturating degree 4 without refutation, or not run. Source: `plot.py` (layout only) on `plot-data.json` from `dreg_second_link_score --plot-data`.

- **Controls:** valid. N matches the committed record, every record passes the identity and exactness checks, and every draw is decided.
- **L3** suffices on 6 of 12 draws, and **S3** on 7 of 12.
- **Discordant draws:** L3-only 0, S3-only 1 (draw 27).
- **Registered verdict:** no detected difference (two-sided exact sign test, p = 1.0).
- **Where both fail:**
  - L3 fails later on draws 0 (b6 against b4) and 14 (b6 against b2).
  - L3 fails earlier on draw 17 (b4 against b6).
  - L3 fails at the same branch on draws 3 and 25.